

EXPLORATIONS LAB AND ACTIVITIES MANUAL



Explorations

Explorations Biological Anthropology Lab and Activities Manual 1e

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Detailed Licensing

Licensing

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CHAPTER OVERVIEW

1: Introduction to Biological Anthropology

Learning Objectives

- Recognize and differentiate scientific fact from different ways of knowing.
- Articulate the importance of differentiating between belief and knowledge without discounting alternative ways of knowing or acquiring beliefs

[1.1: Knowing and Believing](#)

[1.2: Knowing and Believing \(Worksheet\)](#)

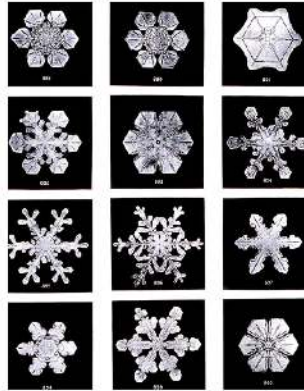
[1.3: Icebreaker Science](#)

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1.1: Knowing and Believing



Snowflakes by [Wilson Bentley](#) (1902). [SnowflakesWilsonBentley](#) by [Wilson Bentley](#) is marked as [public domain](#).

- Format: In-person or online
- Authors: Sydney Quinn Chizmeshya & Katherine E. Brent
- Time needed: 20-40 minutes

Supplies Needed

- Activity sheet for each student (attached), or statements projected on a screen for the class to see
- Writing utensil
- Textbook

Introduction

The learning goal of this activity is for students to become comfortable recognizing and differentiating scientific fact from different ways of knowing. Additionally, students should acknowledge the importance of differentiating between belief and knowledge without discounting alternative ways of knowing or acquiring beliefs in hopes of maintaining cultural relativity.

Procedure

- Distribute one activity page to each student, or project the statements listed below onto a screen. This activity can be done individually, in groups, or as a class. There is also an opportunity to start the activity individually, and to consequently take up answers in a pair-and-share or class discussion.
- Before beginning the activity, students should write the definitions of 'knowledge' and 'belief' in their own words. Subsequently, they should compare these with the definitions put forth in the textbook. Putting class concepts into one's own words helps to solidify concepts for students.
- Next, students should be instructed to read each of the following statements and indicate on the activity sheet, adjacent to each example phrase, whether the statement reflects knowledge or belief.

Statements

1. The sun rises in the East and sets in the West.
 2. All living things need water to survive.
 3. God is real.
 4. No two snowflakes are identical.
 5. Lord of the Flies is the best novel ever written.
 6. Ottawa is the capital of Canada.
 7. The Mona Lisa is the most beautiful painting in the world.
 8. Broccoli is delicious.
 9. Carbon monoxide is dangerous to humans in large quantities.
 10. Dr. G is the best professor.
- After responding to the given statements, each student should write a statement of belief and a statement of knowledge on the activity sheet.

- Once students have labelled each statement accordingly, and written their own statements, discuss in groups, or as a class.
- Ensure that you bring forth any discussions as to why students labelled statements as belief or knowledge into a classroom discussion. Students should use the textbook definitions of “knowledge” and “belief” to aid in justifying their answer.
- Instructors can further engage the class by discussing why it is important for anthropologists to be cognizant of both knowledge and beliefs-- this can introduce the idea of anthropologists as both scientists and cultural relativists.
- Additionally, the idea that knowledge and beliefs can sometimes become ‘tangled’ should be addressed. For instance, it is not uncommon to see scientific knowledge (e.g. that climate change is real) as a belief, or a belief (e.g. that race is biologically founded) as knowledge. As such, differentiation is important.

Statement Answer Key

Statement	Belief or Knowledge?
The sun rises in the East and sets in the West.	Knowledge
All living things need water to survive.	Knowledge
God is real.	Belief
No two snowflakes are identical.	Knowledge
Lord of the Flies is the best novel ever written.	Belief
Ottawa is the capital of Canada.	Knowledge
The Mona Lisa is the most beautiful painting in the world.	Belief
Broccoli is delicious.	Belief
Carbon monoxide is dangerous to humans in large quantities.	Knowledge
Dr. G is the best professor.	Belief
Write your own statement of belief.	<i>To be evaluated</i>
Write your own statement of knowledge.	<i>To be evaluated</i>

Adapting for Online Learning

If this is an in-person lab, rank how adaptable to online learning it would be (mark in bold): **Possible to adapt**

This activity could be using various online programs such as Quizlet, Kahoot, or as an activity completed and submitted through the school’s respective online Learning Management System. Creating an online discussion board to articulate the reasoning behind answers as well as follow up questions is highly recommended. This activity may also be used as an iClicker activity.

References

Nelson, Katie, Lara Braff, Beth Shook, and Kelsie Aguilera. 2019. “Chapter 1: Introduction to Biological Anthropology” In *Explorations: An Open Invitation to Biological Anthropology*, edited by Beth Shook, Katie Nelson, Kelsie Aguilera, and Lara Braff. Arlington, VA: American Anthropological Association. <http://explorations.americananthro.org/>

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1.2: Knowing and Believing (Worksheet)

Step One

In Chapter 1 of *Explorations*, we learned the difference between knowledge and beliefs. Before we begin, answer the following questions in your own words:

1. What is knowledge?
2. What is a belief?
3. What is the difference between these two concepts?

Compare your definitions to those in the textbook.

Step Two

Read the following statements. Using the definition of “belief” and “knowledge” discussed in Chapter 1 of the *Explorations* textbook, fill in the chart to identify which statements are statements of belief or statements of knowledge. Compare results with classroom peers.

Statement	Belief or Knowledge?
The sun rises in the East and sets in the West.	
All living things need water to survive.	
God is real.	
No two snowflakes are identical.	
Lord of the Flies is the best novel ever written.	
Ottawa is the capital of Canada.	
The Mona Lisa is the most beautiful painting in the world.	
Broccoli is delicious.	
Carbon monoxide is dangerous to humans in large quantities.	
Dr. G is the best professor.	

Write Your Own	
Write your own statement of belief.	
Write your own statement of knowledge.	

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1.3: Icebreaker Science

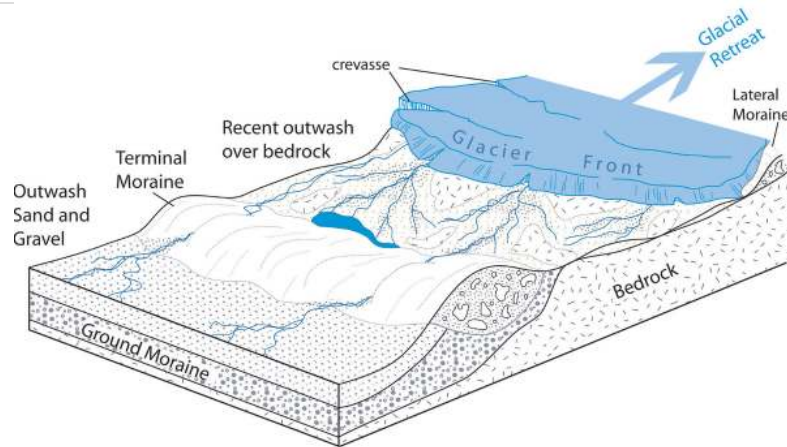


Figure. Illustration of a retreating glacier. Thornberry-Ehrlich, Trish L., Colorado State University. (2017). Glacier Features [Illustration]. United States Public Domain Mark 1.0. <https://flic.kr/p/Rji5E7>.

- Format: In-person
- Author: Keith Chan
- Time needed: 60 minutes

Learning Objectives

- Experience the scientific process
- Interact with classmates
- Explore statistical sampling and analysis

Supplies Needed

- Writing supplies

Readings

- Nelson, Katie, et al. 2019. Chapter 1: Introduction to Biological Anthropology. *Explorations*.

Introduction

The first day of lab sets expectations for the rest of the course. A lot has to be accomplished: attendance, the syllabus, a learning activity, and becoming familiar with each other. This activity addresses the latter two goals in a way that also uses the scientific method and statistical thinking.

Students are given each question ahead of time so they can consider their own answer as well as what the class is like as a group. After this step, students move around the classroom and chat in different groups based on having something in common. When all of the questions have been explored, the instructor can review the results of each item to learn whether the students' expectations were met or not.

Steps

- Ask students to write what percent of the class fit each of the categories listed on the Categories sheet. This is the hypothesis part of the activity.
- For each category, students move to a certain part of the class. For example, students who have been on a roller coaster stand to the left and those who have not stand to the right.
- Give students a minute to meet each other in their group for each category.
- Record the percent of students on each side for each category. This is the observation part of the activity.
- After the categories have all been surveyed, have the class return to their seats.
- Go through each category and ask students for their impressions. Did they get close or not?

Conclusion or Review Questions

This activity got students moving and interacting each other while exploring the scientific process of hypothesis testing. Here are a few questions to conclude the activity:

- Are these results representative of the country? The city? The school? Why or why not? This question can segue into a discussion of bias and sampling methods.
- Which hypotheses could be considered “close enough” to the results? This can lead to an introduction to statistical inference and the standards used in science.

Adapting for Online Learning

If this is an in-person lab, rank how adaptable to online learning it would be (mark in bold): **Possible to adapt**

Students could form breakout rooms for each category, raise their hands, or respond in chat.

Tips and Suggestions

Come up with your own questions related to your school or community. Avoid questions that can be too personal (e.g., are you parents still together or not?) or involve knowledge that is too specific (e.g., did you like the last season of Game of Thrones?).

Icebreaker Science Categories

- Ridden a roller coaster or not
- Is a dog or cat person
- Have been to Europe or not
- Is a morning or night person
- Like to go jogging or not
- Have eaten an insect or not
- Have done karaoke or not
- Have their birthday in the first or second half of the year
- Like to eat seafood or not
- Like group work or solo work

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1.4: Science and Belief- Just Because We Can, Doesn't Always Mean We Should



Figure. Image Caption: International Symbol for Deafness and Hearing Loss. [International Symbol for Deafness and Hearing Loss](#) by the State of Rhode Island is in the public domain.

- Format: In-person or online (discussion or lab activity)
- Author: [Anne E. Pfister, PhD](#)
- Time needed: 45-75 minutes

Learning Objectives

- Distinguish between science and belief
- Apply anthropological theories and methods to understanding real-world issue

Supplies Needed

- Short readings (provided)
- Worksheet (provided)
- Space for discussion (if in-person)
- Discussion board (if online)

Readings

- Nelson, Katie, et al. 2019. Chapter 1: Introduction to Biological Anthropology. *Explorations*
- Sprigs, M. 2002. [Lesbian couple create a child who is deaf like them](#). *Journal of Medical Ethics*. 28:283.
- Ives, James. 2020. [Researchers Treat Deafness with Gene Therapy](#). *News Medical Life Sciences*.

Introduction

Chapter two author, Jonathan Marks (2017) presents anthropology as the “science of mediation”. This lesson and activity invite Marks’s style of intellectual dialogue, encouraging us to find nuanced connections between some seemingly contradictory concepts. In this lesson, we contemplate two different relational concepts fundamental to the anthropological study of human beings as a species.

Students read two articles that approach deafness from different perspectives and analyze the role of culture and biology with regard to human conditions and diversity as well as the role of science and belief in bio-medical intervention (and other scientific pursuits).

Steps

- Introduce (or review) the properties of science and the properties of belief systems. Remind students that both science and belief systems are valid ways of exploring and evaluating phenomena; they are systems we use to understand the world around us. Belief systems take many forms (ethics, morals, philosophy, religion and laws, for example) and science comes from a specific historical context which then spread on a global scale.
- Distribute the worksheet and the readings listed above. Give students approximately 20-30 minutes to read both articles/summaries (they are reading for big ideas, not doing close readings).
- Discuss students’ reactions to the articles.
- Ask students what big ideas are in tension. How can these dyadic concepts be “mediated” through anthropological approaches?
- Dyadic concepts to discuss might include: Culture and biology; science and belief; nature and nurture; ontology and epistemology.

Takeaway Ideas and Points for Review:

All humans use science AND belief systems to explore the world in our attempt to understand it. Sometimes we have the science to ameliorate problems, but our belief system (ethics, morals, laws) dictate whether (and how) we put science into action (or not!). Think about cloning, for example. We have advanced science capable of cloning our pets (see Brogan, 2008 article in the For Further Exploration below). Does that mean we should? What about cloning your little brother?

Therein lies the irony of the presentation of these two articles together. Gene therapy works only for people with genetically inherited deafness (i.e. folks that are born into deaf families). People are born deaf or deafened for many reasons (genetic causes account for about half of childhood cases, [according to the CDC](#)). More importantly, many deaf people, like the couple in one of the articles, do not view deafness as a disability and many people who approach deafness from a deaf cultural perspective do not view deafness as a loss but often as gain (see Bauman et. al., 2014 article in the For Further Exploration below). In fact, some ethnographic research suggests that deaf families (i.e. families with deaf relatives) are the least likely to seek medical intervention (like gene therapy) because they prioritize the shared cultural and linguistic experience in deaf communities and through the use of sign language (see readings in the For Further Exploration section below for examples and discussion).

- Important: Pointing out this irony does not suggest that either approach is wrong. Again, anthropological theories and methods help us embrace that which is apparently contradictory and “mediate” common ground for more holistic understandings of complex issues like deafness.
- A nuanced approach to deafness –one that acknowledges the connections between culture/biology and science/belief – might better focus resources and programs for deaf people and their families.
- Other dyadic concepts to consider: nature & nurture; ontology & epistemology; organic & synthetic; individual & group; extraordinary & mundane

Adapting for Online Learning

If this is an in-person lab, rank how adaptable to online learning it would be (mark in bold): **Easy to adapt**

For Further Exploration

- Bauman, Dirksen and Murray, Joseph. 2014. [An Introduction to Deaf Gain](#). *Psychology Today*.
- Brogan, Jason. 2008. [The Real Reason You Shouldn't Clone Your Dog](#). *Smithsonian Magazine*.
- Centers for Disease Control and Prevention. 2020. [Hearing Loss in Children](#). *National Center on Birth Defects and Developmental Disabilities*.
- Cooper, Rachel. 2007. [Can it be a Good Thing to be Deaf?](#) *The Journal of Medicine and Philosophy*. 32(6):563-583.
- Kusters, Annelies. 2015. *Deaf Space in Adamorobe: An Ethnographic Study in a Village in Ghana*. Gallaudet University Press.
- Mullin, Emily. 2020. [The End of Deafness](#): Gene Therapy Could End Deafness. Should it? *Future Human*.
- Pfister, Anne E. 2017. [Forbidden Signs: Language Socialization and Therapeutic Approaches to Language in Mexico](#). *Ethos*. 45(1):139-161.

References

1. Marks, Jonathan. 2017. *The Alternative Introduction to Biological Anthropology*, 2nd Edition. Oxford University Press.
2. Nelson, Katie, Lara Braff, Beth Shook, and Kelsie Aguilera. 2019. “Chapter 1: Introduction to Biological Anthropology” In *Explorations: An Open Invitation to Biological Anthropology*, edited by Beth Shook, Katie Nelson, Kelsie Aguilera, and Lara Braff. Arlington, VA: American Anthropological Association. <http://explorations.americananthro.org/>

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1.5: Science and Belief (Worksheet)

Embracing Anthropology as a “science of mediation”

Jonathan Marks (2017) presents anthropology as the “science of mediation”. This lesson and activity pursue intellectual dialogue, encouraging us to find nuanced connections between some seemingly-contradictory concepts that inform our understanding of what it means to be human.

In this lesson, we contemplate two different relational concepts fundamental to the anthropological study of human beings as a species.

Step 1

Using the chart below, jot down some adjectives that describe the characteristics of science and belief (you may also use associated words, or those that can be used interchangeably).

SCIENCE	BELIEF

Step 2

Now, do the same for biology and culture. Jot down some adjectives that describe the characteristics of each.

BIOLOGY	CULTURE

Step 3

Now, read the two short articles. Don’t get bogged down by too many details. Instead, read so you understand the general idea behind the two very different approaches to deafness. Then, review the discussion questions below and prepare your thoughts for the discussion. You may wish to jot down your thoughts ahead of the discussion.

Questions for Discussion

1. Prior to reading these articles, did you think about deafness as being primarily biological or cultural?
2. Do either or both articles consider deafness from a predominantly biological approach?
3. Do either or both articles approach deafness from more of a cultural understanding?

4. What are the roles of science and belief in each of the articles?
5. Did reading these articles and discussing them change any of your ideas about science/belief and culture/biology? About approaches to deafness?
6. How can anthropological methods and theories help inform our understandings of deafness and other conditions/statuses/phenomena?
7. What other seemingly-contradictory concepts do these articles underscore?
8. Do you perceive any irony in the contrast between these two articles?

Note

Review the Take-away Ideas Conclusions document once instructed to do so.

Take-away Ideas and Conclusions:

- All humans use science AND belief systems to explore the world and attempt to understand it. Sometimes we have the science to ameliorate problems, but our belief system (ethics, morals, laws) dictate whether (and how) we put science into action (or not!). Think about cloning, for example. We have advanced science capable of cloning our pets (see article below). Does that mean we should? What about cloning your little brother?
- Therein lies the irony of the presentation of these two articles together. Gene therapy works only for people with genetically inherited deafness (i.e. folks that are born into deaf families). People are born deaf or deafened for many reasons (genetic causes account for about half of childhood cases, [according to the CDC](#)). More importantly, many deaf people, like the couple in one of the articles, do not view deafness as a disability and many people who approach deafness from a deaf cultural perspective do not view deafness as a *loss* but often as *gain* (see article below). In fact, some ethnographic research suggests that deaf families (i.e. families with deaf relatives) are the least likely to seek medical intervention (like gene therapy) because they seek the shared cultural and linguistic experience in deaf communities and through the use of sign language (see articles below).
 - **Important:** Pointing out this irony does not suggest that either approach is wrong. Anthropological theories and methods help us embrace that which is apparently contradictory and “mediate” common ground for more holistic understandings of complex issues like deafness.
 - A nuanced approach to deafness –one that acknowledges the connections between culture/biology and science/belief – might better focus resources and programs for deaf people and their families.
- Other dyadic concepts to consider: nature & nurture; ontology & epistemology; organic & synthetic; individual & group; extraordinary & mundane

References

1. Marks, Jonathan. 2017. *The Alternative Introduction to Biological Anthropology*, 2nd Edition. Oxford University Press.

Further Reading

- Bauman, Dirksen and Murray, Joseph. 2014. [An Introduction to Deaf Gain](#). *Psychology Today*.
- Brogan, Jason. 2008. [The Real Reason You Shouldn't Clone Your Dog](#). *Smithsonian Magazine*.
- Centers for Disease Control and Prevention. 2020. [Hearing Loss in Children](#). *National Center on Birth Defects and Developmental Disabilities*.
- Cooper, Rachel. 2007. [Can it be a Good Thing to be Deaf?](#) *The Journal of Medicine and Philosophy*. 32(6):563-583.
- Kusters, Annelies. 2015. *Deaf Space in Adamorobe: An Ethnographic Study in a Village in Ghana*. Gallaudet University Press.
- Mullin, Emily. 2020. [The End of Deafness](#): Gene Therapy Could End Deafness. Should it? *Future Human*.
- Pfister, Anne E. 2019 [Predicament and Pilgrimage: Hearing Families of Deaf Children in Mexico City](#). *Medical Anthropology* 38(3):195-209.
- Pfister, Anne E. 2017. [Forbidden Signs: Language Socialization and Therapeutic Approaches to Language in Mexico](#). *Ethos*. 45(1):139-161.

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CHAPTER OVERVIEW

2: Evolution

Learning Objectives

- Operationalize questions using the scientific method.
- Evaluate methods for scientific study of human phenomena.
- Distinguish between science and dogma.

[2.1: The Scientific Method - Converting Curiosity into Study](#)

[2.2: The Scientific Method - Converting Curiosity into Study \(Worksheet\)](#)

[2.3: Misconceptions about Evolution](#)

[2.4: Misconceptions about Evolution \(Worksheet\)](#)

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2.1: The Scientific Method - Converting Curiosity into Study



The alchemist by Carl Spitzweg is in the public domain

- Format: In-person or online (discussion or lab activity)
- Author: Anne E. Pfister, PhD
- Time needed: 30-40 minutes

Supplies Needed

- Worksheet (provided)
- Videos (links provided)
- Space for discussion (if in-person)
- Discussion board (if online)

Readings

- Nelson, Katie, et al. 2019. Chapter 1: Introduction to Biological Anthropology. *Explorations*
- Marks, Jonathan. 2019. Chapter 2: Evolution. *Explorations*

Introduction

During this activity, students learn to operationalize ideas and form a hypothetical research design using the scientific method. This introductory activity is designed to get students thinking about topics that interest them that fall within the realm of anthropological study. Most of all, it helps them identify what topics can be approached scientifically and which cannot.

Steps

1. Introduce the concepts of science and belief, explaining that humans use both systems to learn about the world. Explain that science is self-correcting while dogma is unwavering. Discuss examples (i.e. we have the science and technology to clone animals, should we clone humans? This illustrates how science depends on ethics, a belief system. Beliefs cannot be scientifically 'tested' while natural phenomena can be).
2. Review the steps of the scientific method and the assumptions of science introduced by Jonathan Marks in the beginning of Chapter Two: Evolution in *Explorations*.
3. View the short video Turtles and Snakes by Mark Rober to underscore how nearly any idea or curiosity can be converted into scientific study using the tenants of the scientific method: https://www.youtube.com/watch?feature=player_embedded&v=k-Ep7fIAWMA#!
4. Distribute worksheets and have students work independently (or in pairs).
5. (Optional) ask students to share the research topics they brainstormed for the class discussion.
6. Discuss take-away ideas with the class.

Takeaway Ideas and Points for Review:

- All humans use science and belief systems to explore the world and create theories about it.
- Science is a specific way of investigating the world. Scientists use the scientific method because it helps us operationalize ideas and standardize approaches to answering difficult questions and creating theories about our observations.

- Science is self-correcting – as new (and potentially contradictory) data are introduced, theories are amended or improved upon. Dogma is static and does not change.

Adapting for Online Learning

If this is an in-person lab, rank how adaptable to online learning it would be (mark in bold): **Easy to adapt**

Tips and Suggestions

This lab activity and discussion help students and professors identify the interests of individuals in the class. It also motivates students to know that virtually anything they are curious about can be operationalized into a research project using the scientific method.

For Further Exploration

- The Scientific Method, Sprouts: <https://www.youtube.com/watch?v=yi0hwFDQTSQ&v=1=en>
- Science vs. Dogma, Nature: <https://www.nature.com/articles/23307>

References

1. Nelson, Katie, Lara Braff, Beth Shook, and Kelsie Aguilera. 2019. "Chapter 1: Introduction to Biological Anthropology" In *Explorations: An Open Invitation to Biological Anthropology*, edited by Beth Shook, Katie Nelson, Kelsie Aguilera, and Lara Braff. Arlington, VA: American Anthropological Association. <http://explorations.americananthro.org/>
2. Marks, Jonathan. 2019. "Evolution" In *Explorations: An Open Invitation to Biological Anthropology*, edited by Beth Shook, Katie Nelson, Kelsie Aguilera, and Lara Braff. Arlington, VA: American Anthropological Association. <http://explorations.americananthro.org/>

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